CS-013 Work Orders & Part Traceability

Doc ID	CS-013
Title	Work Orders & Part Traceability
Revision	A
Status	Draft — pending Lead signature
Effective date	2026-07-12
Supersedes	Master Tracker part rows as the only part record

Approvals

Role	Name	Date
Author	Simon Starbuck (drafted w/ Claude — SN6 Resources)	2026-07-12
Reviewer	simon (reviewer agent)	
Approver (Composites Lead)		

Revision history

Rev	Date	Author	Description of change
A	2026-07-12	Claude	Initial issue — defines the WO record and its authority rules

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1. Purpose

The Master Tracker did its job at the fleet level — parts, owners, deadlines, all visible. What it never gave us was the *part-level* record: steps, buy-offs, the stack as actually laid, measurements, photos. All of that lived in Slack threads and people's heads (PP-09). A work order (WO) is one part's complete record — spec, blockers, execution, evidence — on a screen or printed and kept with the part.

2. Scope

Every manufactured composites part and mold, including small cross-team jobs (the catch-can class of work — PP-05) and R&D coupons feeding decisions. If it consumes carbon or machine time, it gets a WO.

3. References

Ref	Document	Where
R1	PP-09, PP-05, PP-03 RCAs	FEB-COMP-PP-001
R2	Work-order app + JSON schema	`03 Work Orders/work-orders.html`, `03 Work Orders/README.md`
R3	CS-002 (stack), CS-003 §7.2 (design review), CS-006/007 (process records)	this series

Ref	Document	Where
R4	CS-000 (revision rules apply to WOs)	CS-000

4. Definitions

- **WO ID** — `WO-SNx-###`, assigned from the app (or the next free number in the printed log). Never reused.
- **Buy-off** — typed initials + date on a step by the person who *verified* it (self-buy-off allowed except where a step names an independent reviewer, e.g. mold design review).
- **Blocker step** — a step that blocks everything after it until it's bought off (stack freeze, design review, drop test).
- **Retro WO** — a back-filled historical record (all `WO-SN5-*`); its unverifiable fields say "not recorded (retro)" — retro records are honest references, not exemplars of process compliance.

5. Equipment & Materials

The WO app (R2): a single HTML file, no install, runs in any browser from Drive/local copy; data autosaves in the browser and exports/imports as JSON. Printed WOs: any printer + a pen at RFS.

6. Safety

No physical hazards — process standard. (Safety content lives in the process standards each WO step cites.)

7. Procedure

7.1 Create

1. Open the app → New Work Order → it assigns the next `WO-SN6-###`.
2. Fill at creation (this is the PP-04/PP-05 fix — requirements exist before work): part name, subteam, ME/RE, due date, process type, **stack (per CS-002)**, BOM, **qualityChecks with numeric targets**, and the standard step list for the process type (the app pre-fills steps from the templates; delete inapplicable ones consciously, don't skip silently).
3. Cross-team parts: the requesting subteam's requirements (thickness, DXF rev, mass) go in the WO and their lead is named on the release sign-off.

7.2 Execute

1. Work the steps in order; blocker steps block until bought off (the app enforces this; on paper the blocker rows are shaded — no initials, no moving on).
2. Each step: status, initials + date, notes, photo references (phone photos → the part's Drive folder, filename written on the WO).
3. Deviations from the frozen stack/spec = WO revision (CS-000 rules): quick, visible, counted. The revision letter rides the WO header.
4. Update `mold.location` whenever the mold physically moves (PP-10).

7.3 Close

1. Fill qualityChecks actuals (mass, thickness, Ω table, etc.), attach final photos, mark status Complete.
2. Export JSON (the app's export button) into the season's Drive folder — **the JSON export is the durable archive; the browser's local storage is a cache, not a record.**
3. Print one copy for the physical binder at RFS if the part is safety/inspection-relevant (mech tech evidence).

7.4 Weekly use

1. Monday meeting: the app's list view (filter: InWork) *is* the part status board — it replaces re-asking "where is X?" and feeds the weekly summary.
2. The lead exports all JSON weekly (one click) — that snapshot is the season's rolling backup.

8. Quality checks / acceptance criteria

Check	Target	Method	Record where
Every active part has a WO	100% of parts consuming carbon/machine time	Lead check at Monday meeting	—
Blocker steps bought off in order	No downstream work past an unfinished blocker	App enforcement / shaded rows on paper	WO
Weekly JSON export	Every week during build season	Lead	Drive folder timestamps
Retro honesty	No fabricated buy-offs on retro WOs	"not recorded (retro)" convention	WO fields

9. Records

The WO JSON archive in Drive is the team's part-history database. The Master-Tracker-style views (spreadsheets, summaries) may continue — as *views*. The WO is the source of truth (same rule as CS-002 §9).